CO-3_Assessment Tool-1

EXPERIMENTS

CSA0712-Computer Networks

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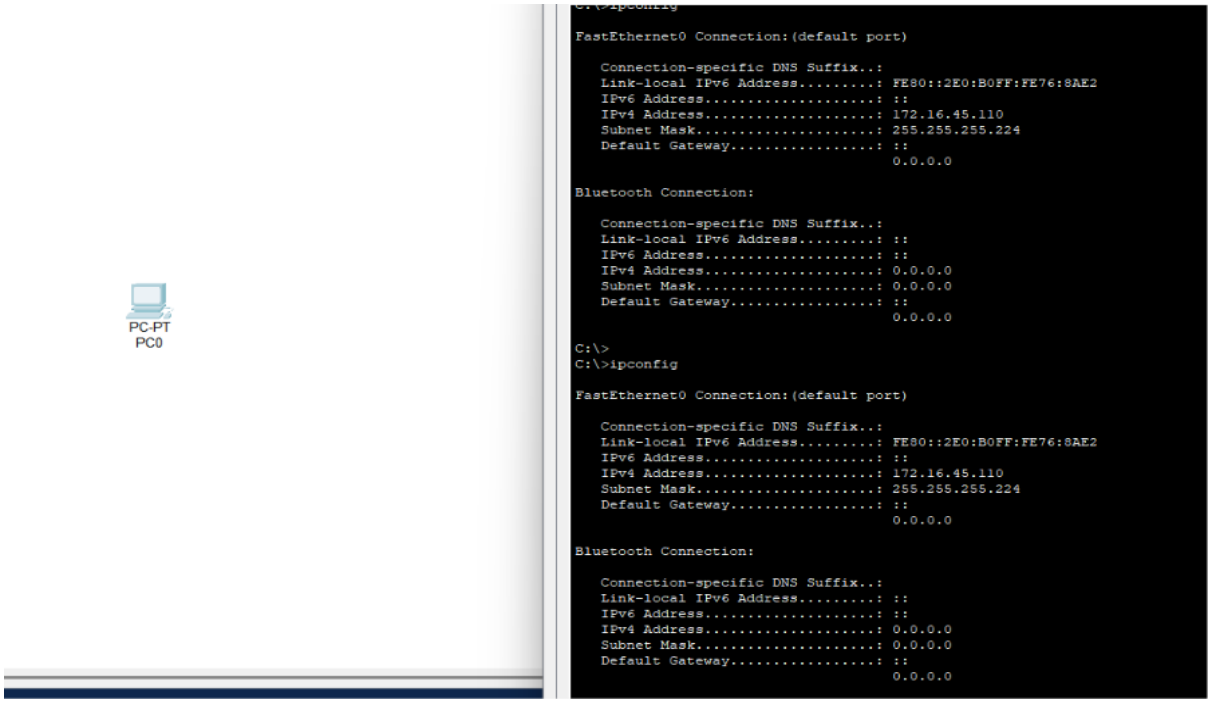
Part 1

Experiment 2: Subnet Calculation from Host IP

Scenario: During a Wireshark capture experiment, a student notices a workstation sending packets from the IP address 172.16.45.110 with a subnet mask of 255.255.255.224.

Tasks:Calculate the exact Network Address to which this workstation belongs.Determine the First Usable IP and the Last Usable IP address in this specific subnet.Provide the Broadcast Address for this subnet block.

Answer:



256 - 224 = 32

110 lies between 96 and 127

Network Address = 172.16.45.96

First Usable = 172.16.45.97

Last Usable = 172.16.45.126

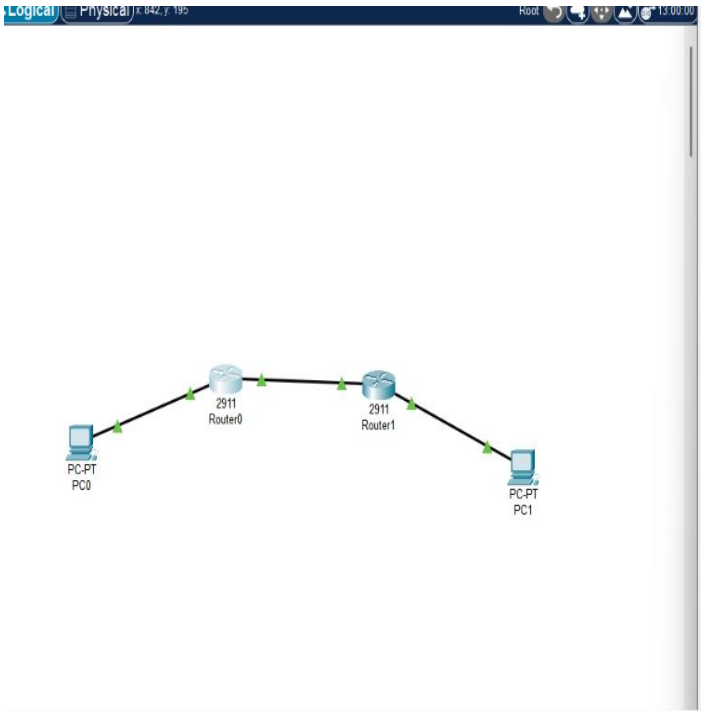
Broadcast = 172.16.45.127

Part-2

Experiment 3: Cisco Command Line Interface (CLI) Static Routing
Scenario: You are setting up a laboratory topology in Cisco Packet Tracer consisting of two routers (Router_A and Router_B) connected via their GigabitEthernet0/0 interfaces.

- Router_A is connected to Local LAN 1: 192.168.1.0/24
- Router_B is connected to Local LAN 2: 192.168.2.0/24
- The link between the two routers uses the subnet 10.10.10.0/30. Router_A holds 10.10.10.1 and Router_B holds 10.10.10.2.

Tasks:Write out the exact global configuration mode commands required to configure a manual static route on Router_A so it can successfully reach LAN 2.Write the command to configure a default static route on Router_B that points all unknown traffic back toward Router_A.



Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.10.10.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0
L       10.10.10.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
S    192.168.2.0/24 [1/0] via 10.10.10.2

Router#show ip route static
S    192.168.2.0/24 [1/0] via 10.10.10.2
```

Part-3

Experiment 6: Network Troubleshooting & Path Discovery

Scenario: A lab technician runs a traceroute 8.8.8.8 command from a desktop computer. The output shows asterisks (* * *) starting from the third hop onward, and terminal communication completely fails.

Tasks:Explain the underlying network layer protocol and the specific header field manipulated by the traceroute utility to map out hops.Diagnose what the asterisks or a sudden drop-off typically indicates in terms of router behavior or firewall security policies at the third hop.

Answer:

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harsha>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=37ms TTL=116
Reply from 8.8.8.8: bytes=32 time=327ms TTL=116
Reply from 8.8.8.8: bytes=32 time=41ms TTL=116
Reply from 8.8.8.8: bytes=32 time=31ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 31ms, Maximum = 327ms, Average = 109ms

C:\Users\harsha>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    17 ms     *       52 ms  LAPTOP-ICTLU29T.mshome.net [192.168.137.1]
  2  LAPTOP-ICTLU29T.mshome.net [192.168.137.1]  reports: Destination net unreachable.

Trace complete.

C:\Users\harsha>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

```
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : mshome.net
   Link-local IPv6 Address . . . . . : fe80::2795:c301:ecce:1777%9
   IPv4 Address. . . . . . . . . . . : 192.168.137.46
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.137.1

C:\Users\harsha>tracert -d 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops

  1    59 ms     *        6 ms  192.168.137.1
  2  192.168.137.1  reports: Destination net unreachable.

Trace complete.

C:\Users\harsha>tracert 1.1.1.1

Tracing route to one.one.one.one [1.1.1.1]
over a maximum of 30 hops:

  1    19 ms     *       96 ms  LAPTOP-ICTLU29T.mshome.net [192.168.137.1]
  2  LAPTOP-ICTLU29T.mshome.net [192.168.137.1]  reports: Destination net unreachable.

Trace complete.

C:\Users\harsha>
```